

1. Administrative & Study Identification

Official Regulatory Title: A Phase 2a Randomised, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of AZD4604 Twice Daily for Twelve Weeks in Adult Patients with Moderate-to-Severe Asthma Uncontrolled on Medium-High Dose ICS-LABA **Study Title:** AJAX **Study Primary ID:** D8210C00003 **Registry Number:** NCT06020014 **Sponsor/Lead Organization:** AstraZeneca **Investigational Product:** AZD4604 (londamocitinib), an inhaled JAK1 inhibitor (Dry-powder inhaler) **Primary Condition:** Moderate-to-Severe Asthma **Phase of Development:** Phase IIa **Medical Monitor:** Emmanuel Amadi, AstraZeneca Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of AZD4604 administered twice daily (BID) compared to placebo in adult participants with moderate-to-severe asthma by assessing the time to the first CompEx Asthma event over a 12-week treatment period. **Secondary Objectives:** To evaluate the effect of AZD4604 on lung function (change in pre-bronchodilator FEV1), asthma control and symptoms (CAAT, ACQ-6, daily symptom score, morning/evening PEF), airway inflammation (FeNO), cough severity, and the overall safety, tolerability, and pharmacokinetics (PK) of AZD4604.

2.2 Background & Rationale To address the unmet medical need in patients with moderate-to-severe asthma whose symptoms remain uncontrolled despite medium-high doses of inhaled corticosteroids and long-acting beta-agonists (ICS-LABA). While oral Janus kinase (JAK) inhibitors are used for other inflammatory diseases, they can have systemic side effects. The **Medical Methodology** for this trial focuses on **Targeted JAK1 Inhibition (londamocitinib)**. AZD4604 is an inhaled JAK1 inhibitor specifically targeting airway inflammation. The **Optimization Target** of this intervention is the **Reduction of airway inflammation and asthma exacerbations**, potentially offering improved asthma control—especially in non-Type 2 (non-eosinophilic) asthma phenotypes—without the widespread systemic effects of oral medications.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target **\$N\$**: Approximately 320 participants **Number of Sites**: ~150 sites across 18 countries
Total Duration: 12 Weeks Treatment + Post-treatment Follow-up
Estimated Study Start: 16-Nov-2023 **Estimated Primary Completion**: 28-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult men or women 18 to 80 years of age. 2. Documented clinical diagnosis of asthma at least 12 months prior to Screening. 3. **Diagnosis Threshold**: Uncontrolled on Medium-High Dose ICS-LABA, ACQ-6 ≥ 1.5 , pre-BD FEV1 $\geq 40\%$ predicted at Screening and Baseline. 4. Documented history of ≥ 1 severe asthma exacerbation within 1 year prior to Visit 1.

4.2 Exclusion Criteria 1. History of malignancy other than superficial basal cell carcinoma. 2. Treatment with systemic corticosteroids within 4 weeks (oral) or 8 weeks (intramuscular) before Visit 1, or any immunosuppressive therapy within 12 weeks prior to Visit 1. 3. Treatment with marketed biologics within 6 months of Visit 1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: AZD4604 or matching placebo administered twice daily (BID). **Route of Administration**: Oral inhalation via a dry-powder inhaler. **Management Pathway**: 12-week Double-Blind Treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background asthma maintenance therapy (medium-high dose ICS-LABA). **Prohibited**: Systemic corticosteroids, immunosuppressives, marketed biologics, and the use of ICS plus fast-acting β_2 agonist as a reliever 15 days prior to Visit 1 and throughout the treatment period.

6. Study Timeline & Visit Schedule

Onsite Visit Cadence: Screening, Baseline, Week 4, Week 12

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Lung Function (FEV1), ACQ-6 Assessment, Background Medication Stability Check
Baseline	Day 0	Randomization, First Dose, Pre-BD FEV1, CAAT, ePRO Device Dispensation	Interim
Week 4	$\pm$ [Y]	Safety Labs, Efficacy Assessment (FEV1, ACQ-6, CAAT), FeNO, PK Sampling	End of Study
Week 12	Final Lung Function, Cough Severity, FeNO, Exit Interview, IP Accountability		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to first CompEx Asthma acute worsening event over the 12-week period. **Measurement Method:** A composite surrogate endpoint based on electronic Patient-Reported Outcomes (ePRO) tracking asthma symptoms and rescue medication use, combined with Peak Expiratory Flow (PEF) data and clinical severe asthma exacerbation events.

7.2 Secondary & Exploratory Endpoints * Change from baseline in pre-bronchodilator forced expiratory volume in 1 second (FEV1) at Weeks 4 and 12. * Change from baseline in the Chronic Airways Assessment Test (CAAT) and Asthma Control Questionnaire (ACQ-6). * Change from baseline in fractional exhaled nitric oxide (FeNO) levels. * **Checklist Review Interval:** Bi-weekly (14-day) average analysis of symptom scores and PEF logs.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via site visits, HCP interviews, and a **Remote Monitoring Frequency of Continuous daily ePRO/PEF monitoring.** **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline to the sponsor and applicable regulatory authorities. **Data Monitoring Committee (DMC):** External and internal safety review boards governed by AstraZeneca to monitor double-blind trial safety events.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for all primary and secondary efficacy endpoints; Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** $N = 320$ properly powered ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect a statistically significant reduction in the hazard ratio for the time to the first CompEx Asthma event compared to placebo. **Handling of Missing Data:** Multiple Imputation for missing continuous variables (e.g., FEV1, ACQ-6) and specific censoring rules applied to time-to-event survival analyses for the primary endpoint.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Education Protocol (HCP):** Training on CompEx Asthma endpoint tracking and centralized spirometry parameters. **Education Protocol (Patient):** Inhaler technique training and daily ePRO symptom/PEF diary completion. **Monitoring Plan & Quality Audit**

Mechanism: Onsite monitoring visits supplemented by a robust electronic data capture (EDC) system enabling **ePRO compliance monitoring & centralized lung function review**, alongside automated eCRF locking and rigorous data clarification forms.

11. Study Identifiers & Governance

Primary ID: {{D8210C00003}} **Registry Number:** {{NCT06020014}}
Sponsor/Lead Organization: {{AstraZeneca}} **Study Title:** {{AJAX Study}} **Official Regulatory Title:** {{A Phase 2a Randomised, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of AZD4604 Twice Daily for Twelve Weeks in Adult Patients with Moderate-to-Severe Asthma Uncontrolled on Medium-High Dose ICS-LABA}}

12. Clinical Framework (Asthma Specific)

Primary Condition: {{Moderate-to-Severe Asthma}} **Diagnosis Threshold:** {{Uncontrolled on Medium-High Dose ICS-LABA, ACQ-6 \$ \ge 1.5\$, pre-BD FEV1 \$ \ge 40\%\$}} **Medical Methodology:** {{Targeted JAK1 Inhibition (londamocitinib)}} **Optimization Target:** {{Reduction of airway inflammation and asthma exacerbations}}

13. Study Procedures & Methodology

Management Pathway: {{12-week Double-Blind Treatment}} **Education Protocol (HCP):** {{Training on CompEx Asthma endpoint tracking and centralized spirometry parameters}} **Education Protocol (Patient):** {{Inhaler technique training and daily ePRO symptom/PEF diary completion}} **Quality Audit Mechanism:** {{ePRO compliance monitoring & centralized lung function review}}

14. Observation & Follow-Up Schedule

Total Duration: {{12 Weeks Treatment + Post-treatment Follow-up}}
Onsite Visit Cadence: {{Screening, Baseline, Week 4, Week 12}}
Remote Monitoring Frequency: {{Continuous daily ePRO/PEF monitoring}} **Checklist Review Interval:** {{Bi-weekly (14-day) average analysis of symptom scores and PEF logs}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				